

EMEL ÖRKEN

Backend Developer | Python & Django | Microservices | Distributed Systems

+90 505 232 42 13 | orkenemel.13@gmail.com | linkedin.com/in/emel-örken-ab229a274

<https://github.com/orken13> | <https://orken13.github.io/emel-portfolio/>

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with proven experience designing and delivering distributed microservice systems in Python. Built a production 3-container Docker architecture (RabbitMQ + PostgreSQL + MinIO) with async message pipelines and a live REST API — independently, from system design to deployment. Proficient in Python, Flask/FastAPI, PostgreSQL, Redis, and RabbitMQ. Strong analytical mindset, high ownership mentality, and a fast learner who thrives in high-paced engineering environments.

EXPERIENCE

Software Engineer — ML & Backend Systems

Oct 2025 – Feb 2026 | Remote

TURK AI

- Engineered a fully automated Python data pipeline: field data ingestion → augmentation → model training → COCO metric evaluation → structured inference reporting — reducing manual overhead and ensuring reproducibility across runs.
- Took full post-delivery ownership of ML pipeline infrastructure: monitored performance metrics, resolved data quality issues, and continuously improved the system across multiple training iterations.
- Managed and quality-controlled a dataset of 5,000+ real-world images (7 classes); enforced annotation standards and resolved inconsistencies to maintain reliable, production-grade training data.

Stack: Python · CVAT · PyTorch · YOLOv11n/12n · OpenCV · Roboflow · Kaggle · Hugging Face

AI Engineer

Jul 2025 – Aug 2025

MANK Technologies

- Built a large-scale audio data processing pipeline in Python: automated ingestion, format conversion (WAV + SRT), and text normalization (number-to-word) — processing 9+ hours of data with zero manual intervention.
- Designed and executed a model benchmarking framework evaluating three ASR systems (Whisper Large v3, Whisper Turbo, Phi-4) against 2.32h of test data using WER/CER metrics; delivered a structured technical report to the team lead.
- Maintained reproducible training configurations across multiple GPU environments (H100, A100, RTX 5090); documented all experimental results and contributed to R&D decisions through data-driven analysis.
- Contributed to cross-functional R&D by rapidly prototyping and evaluating new AI technologies; delivered a working voice-based customer service demo within two weeks of assignment.

Stack: Python · PyTorch · Whisper · Ffmpeg · Git · H100 / A100 / RTX 5090

Backend Engineer Intern

Feb 2025 – Jun 2025 | Ankara

TURK AI

- Architected and deployed a 3-container Docker microservice system (RabbitMQ async message queue + PostgreSQL + MinIO object storage) for real-time data tracking; built a Flask REST API with live dashboard, filtering, sorting, and pagination — delivered end-to-end from design to production.
- Built a full backend data pipeline: automated transcription (Whisper) → vector embedding → multi-collection semantic search API — implemented with clean, modular Python abstractions and took ownership of ongoing improvements.

Stack: Python · Docker · RabbitMQ · PostgreSQL · MinIO · Flask · Redis · WebSocket · LangChain · Git

Backend Engineer Intern

Sep 2024

CBOT

- Built a document-based question answering backend using FAISS and LangChain; designed clean Python abstractions for chunking, embedding, and retrieval — maintainable, testable code from day one.
- Prototyped multiple AI-powered backend services (multimodal, speech processing) and iterated rapidly based on stakeholder feedback using Gradio and Streamlit interfaces.

- Delivered multiple working prototypes under tight deadlines; gained hands-on experience with professional code standards, version control (Git/GitHub), and cross-team collaboration.

Stack: LangChain · FAISS · CLIP · BLIP · Whisper · Tortoise TTS · Gradio · Streamlit .LLM .HuggingFace

NOTABLE PROJECTS

INTERPOL Real-Time Data Tracking System

3-container Docker microservice architecture · RabbitMQ async message queue · PostgreSQL · Redis caching · MinIO object storage · Flask REST API with live dashboard — designed, built, and owned end-to-end.

Turkish TTS Model Training — 70+ Hours of Audio

XTTS trained on H100 with 30 speakers · MFA phonetic alignment · text normalization pipeline · WER/CER benchmark report — production-grade speech synthesis system.

Production Object Detection System (TURK AI)

Real field images · YOLOv11n/12n training · Roboflow augmentation and annotation of 5,000+ images in CVAT · systematic COCO metric evaluation · RTX 5090 GPU infrastructure.

YouTube RAG Chatbot

Python backend pipeline: Whisper transcription → chunking → vector embedding → multi-collection semantic search API — modular, maintainable architecture with full post-delivery ownership.

AI Agent Backend System — FastAPI + LangGraph

FastAPI + LangGraph multi-agent backend · FAISS RAG · two-layer memory (SQLite + sliding window) · Redis · WebSocket server · local LLM via Ollama — fully async, production-grade agentic system.

TECHNICAL SKILLS

Languages:	Python · SQL · Go (basic) · C#
Deep Learning:	PyTorch · TensorFlow · Transformers · Hugging Face
Computer Vision:	YOLOv11/v12 · OpenCV · Roboflow · CVAT
Speech AI:	VITS · XTTS · Whisper (Large v3 / Turbo) · MFA · Tortoise TTS
Generative AI:	LangChain · LangGraph · RAG · FAISS · Vector Databases · Ollama
Backend & Infra:	Python · Django · Flask · FastAPI · PostgreSQL · Redis · RabbitMQ · Apache Kafka · Docker · WebSocket · MinIO · Git
GPU Platforms:	H100 · A100 · RTX 5090 · RTX 4090
Tools:	Kaggle · Gradio · Streamlit · Diffusers

EDUCATION

B.Sc. Software Engineering

Atatürk University, Erzurum, Turkey | GPA: 3.26 / 4.00

Sep 2021 – Jun 2025

Selçuklu Science High School

GPA: 95.2 / 100 | Ranked 2nd in school

Oct 2016 – Jun 2020

AWARDS & ADDITIONAL

Awards: TÜBİTAK High School Research Projects — Regional Acceptance (2019) · TEV Full Scholarship (2022–2025)

Volunteer: Türk Eğitim Vakfı (TEV) Menti Program

Reference: Barış Ünver · +90 505 820 23 03 (Turk AI Founder & CEO)

Reference: Tüncer Keleş · +90 532 603 49 93 (İK Director)

Reference: Ersin Aksoy · +90 505 245 91 81 (Lotus AI Founder & CEO)

Reference: Ceyda Süer · +90 542 561 10 03 (Flexvision Tech Founder & CEO)
